

Exploratory Data Analysis of the Spam Email Dataset

1. Introduction

Exploratory Data Analysis (EDA) was performed to understand the structure, quality, distribution, and characteristics of the spam email dataset before applying machine-learning algorithms. The analysis helps identify missing values, duplicate records, class imbalance, text-length characteristics, and potential outliers.

The dataset is used for a supervised binary classification problem in which emails are classified into two categories: legitimate (ham) emails and spam emails.

2. Dataset Overview

The original dataset contains **6,048 email records and 2 columns**:

Column	Data Type	Description	Role
<code>label</code>	int64	Class assigned to each email	Target variable
<code>origin</code>	object	Raw email content	Input feature

The `label` column contains two unique values, **0 and 1**, making this a binary classification problem.

For this project:

- **Input feature (X):** `origin`
- **Target variable (y):** `label`
- **Learning type:** Supervised Machine Learning
- **Problem type:** Binary Classification

The `origin` column contains the raw email content, including email headers and message information such as `Return-Path`, `Delivered-To`, and `Received` fields. This textual information will later be processed and converted into numerical features using TF-IDF.

During EDA, an additional column called `email_length` was created to measure the number of characters in each email. Therefore, the DataFrame contains **3 columns after EDA**, although the original dataset contains only 2 columns.

3. Dataset Structure and Data Types

The dataset contains **6,048 entries**, indexed from 0 to 6047.

The data types are:

- `label` : integer (`int64`)
- `origin` : object/string (`object`)

There are **6,048 non-null values in both original columns**, indicating that every record contains both a class label and email content.

The dataset occupies approximately **94.6 KB** of memory for the two original columns.

Observation

The data types are appropriate for the classification task. The `label` column is numerical and can directly represent the target class, while the `origin` column contains textual data that can later be processed using Natural Language Processing techniques.

4. Missing Value Analysis

Missing values were examined for all columns.

Column	Missing Values	Missing Percentage
<code>label</code>	0	0.00%
<code>origin</code>	0	0.00%

The dataset contains a total of **0 missing values**.

Observation

Since neither the target variable nor the email-content column contains missing values, no missing-value imputation is required. This provides a clean starting point for the subsequent text-processing and machine-learning stages.

5. Duplicate Data Analysis

Duplicate records were checked using the `duplicated()` function.

The analysis found:

- **Duplicate records:** 0
- **Duplicate percentage:** 0.00%

Observation

There are no duplicate rows in the dataset. Therefore, no duplicate records need to be removed before proceeding with the machine-learning pipeline.

6. Target Variable Analysis

The target variable is `label`.

The dataset contains two classes:

- **0:** Ham / legitimate email
- **1:** Spam email

The target distribution is shown below.

Class	Number of Emails	Percentage
0 — Ham	4,151	68.63%
1 — Spam	1,897	31.37%
Total	6,048	100%

Observation

The dataset contains significantly more legitimate emails than spam emails. Ham emails represent **68.63%** of the dataset, while spam emails represent **31.37%**.

This indicates that the dataset is **moderately imbalanced** toward the ham class.

This class distribution is important when evaluating the classification models. Accuracy alone may not provide a complete representation of model performance. Therefore, additional evaluation metrics such as **precision, recall, F1-score, and confusion matrix** will be used in later stages.

7. Email Content Analysis

The `origin` column contains the complete raw email information. A check was performed to determine whether any email content was missing.

Missing email contents: 0

Therefore, all 6,048 records contain email text.

Sample records show that the dataset contains email headers and message-related information, including fields such as:

- Return-Path
- Delivered-To
- Received
- Email addresses
- Other email metadata

Observation

The presence of raw textual email content makes the dataset suitable for a text-classification approach. Instead of using manually extracted numerical features, the project can apply text preprocessing followed by TF-IDF feature extraction.

8. Email Length Analysis

An additional EDA feature called `email_length` was created by calculating the number of characters in each email.

The descriptive statistics are:

Statistic	Value
Count	6,048
Mean	5,346.9977
Standard Deviation	9,207.2625
Minimum	368
25th Percentile (Q1)	2,505
Median	3,454
75th Percentile (Q3)	4,926.5
Maximum	300,736

The average email length is approximately **5,347 characters**, while the median email length is **3,454 characters**.

The large difference between the mean and median indicates that the email-length distribution is **right-skewed**, with some emails being substantially longer than the majority of records.

The maximum observed email length is **300,736 characters**, which is considerably larger than the median.

9. Email Length by Class

The average and median email lengths were calculated separately for ham and spam emails.

Average Email Length

Class	Average Length
Ham (0)	4,801.32 characters
Spam (1)	6,541.05 characters

Median Email Length

Class	Median Length
Ham (0)	3,336 characters
Spam (1)	4,037 characters

Observation

Spam emails have a greater average and median length than ham emails in this dataset.

The average spam email length is approximately **6,541 characters**, compared with approximately **4,801 characters** for ham emails.

Similarly, the median spam email length is **4,037 characters**, while the median ham email length is **3,336 characters**.

This suggests that email length may contain some useful information for distinguishing the two classes. However, email length alone will not be used as the primary classification feature. The complete email text will later be transformed into TF-IDF features.

10. Outlier Detection

Potential outliers were identified using the **Interquartile Range (IQR)** method.

The calculated values were:

- **Q1 = 2,505**
- **Q3 = 4,926.5**
- **IQR = 2,421.5**

The lower and upper bounds were calculated as:

$$\text{Lower Bound} = Q1 - 1.5 \times IQR$$

$$\text{Lower Bound} = 2,505 - 1.5 \times 2,421.5 = -1,127.25$$

$$\text{Upper Bound} = Q3 + 1.5 \times IQR$$

$$\text{Upper Bound} = 4,926.5 + 1.5 \times 2,421.5 = 8,558.75$$

Therefore, emails with lengths greater than **8,558.75 characters** were identified as potential outliers.

Outlier Results

- **Potential outliers:** 653
- **Outlier percentage:** 10.80%

The outliers were distributed as follows:

Class	Potential Outliers
Ham (0)	295
Spam (1)	358
Total	653

Observation

Approximately **10.8%** of the emails were identified as potential outliers based on their length.

There are more potential outliers among spam emails (**358**) than ham emails (**295**).

However, these observations should **not automatically be removed**. A very long email can be a legitimate email or a genuine spam example. Removing such records without investigation could result in the loss of useful information.

Therefore, the outlier analysis is used primarily to understand the dataset rather than automatically delete these records.

11. Visualization Analysis

Several visualizations were generated during EDA.

11.1 Spam vs Ham Distribution

A count plot was generated to visualize the target-class distribution.

The graph confirms that ham emails are more numerous than spam emails:

- Ham: **4,151**
- Spam: **1,897**

This visualization clearly demonstrates the moderate class imbalance in the dataset.

11.2 Email Length Boxplot

A boxplot was used to visualize the distribution of email lengths for the two classes.

The boxplot indicates the presence of several high-length observations, particularly beyond the upper IQR boundary of **8,558.75 characters**.

11.3 Email Length Distribution

A histogram was generated to examine the distribution of email lengths.

The distribution is strongly right-skewed because a relatively small number of emails have very large lengths compared with the majority of the dataset.

12. Sample Email Inspection

Sample emails from both classes were inspected to understand the raw data.

Examples of ham emails included messages with email headers such as:

- Return-Path: <fork-admin@xent.com>
- Return-Path: <bounce-lglinux-...>
- Return-Path: <ilug-admin@linux.ie>

Examples of spam emails included messages containing addresses and headers such as:

- Return-Path: <batonbrat6@hotmail.com>
- Return-Path: <FLN-Community@FreeLinksNetwork...>
- Return-Path: <crackmice-admin@crackmice.com>

Random samples also confirmed that the dataset contains both ham and spam records throughout the dataset.

Observation

The samples demonstrate that the `origin` column contains raw email information rather than pre-extracted numerical features. This supports the use of an NLP-based approach in the next stage.

13. Overall EDA Findings

The exploratory analysis produced the following key findings:

EDA Component	Finding
Total records	6,048
Original columns	2
Input feature	<code>origin</code>
Target variable	<code>label</code>
Classification type	Binary
Learning type	Supervised
Missing values	0
Duplicate records	0
Ham emails	4,151 (68.63%)
Spam emails	1,897 (31.37%)
Mean email length	5,346.9977 characters
Median email length	3,454 characters
Minimum email length	368 characters
Maximum email length	300,736 characters
Potential outliers	653 (10.80%)
Ham outliers	295
Spam outliers	358

14. Conclusion of EDA

The exploratory data analysis shows that the dataset is suitable for developing a supervised machine-learning model for spam email classification.

The dataset contains **6,048 complete email records with no missing values and no duplicate rows**. The target variable contains two classes, with **4,151 ham emails (68.63%)** and **1,897 spam emails (31.37%)**.

The email-length analysis indicates that spam emails tend to be longer on average than ham emails in this dataset. The distribution is right-skewed, with **653 potential outliers**, representing **10.8%** of the dataset. These outliers were identified but were not automatically removed because unusually long emails may contain meaningful information for spam classification.

Overall, the EDA confirms that the dataset contains raw textual email information suitable for Natural Language Processing. The next stage of the project will therefore focus on **text preprocessing and TF-IDF feature extraction**, followed by training and comparing **Decision Tree and XGBoost classifiers using 5-fold cross-validation**.